

# Johanna Loyer

*Postdoctoral Researcher in Cryptology  
and Quantum Computing*

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*My research interests focus on the **cryptanalysis** of **lattice**-based and **code**-based schemes, through the analysis of both classical and **quantum** attacks. I am particularly interested in transfers between these domains.*

## Professional Experience

- since Feb. 2025 **Postdoctoral Researcher**, *Inria Saclay*, Palaiseau, GRACE team  
with Thomas Debris-Alazard
- Feb. 2024 – Jan. 2025 **Postdoctoral Researcher**, *Centrum Wiskunde & Informatica (CWI)*, Amsterdam  
(Netherlands), Cryptology group  
with Léo Ducas
- Sept. 2020 – Dec. 2023 **PhD in Computer Science**, *Inria*, Paris, COSMIQ team  
Thesis title: *Quantum Cryptanalysis on Lattices and Codes*  
advised by André Chailloux and Nicolas Sendrier

## Teaching

- 2020–2023 **Teaching Assistant**, *Sorbonne University*, Paris
- Tutorials: Quantum circuits and logic gates, Master 1 in Quantum Information
  - Tutorials: Quantum information overview, Master 1 in Quantum Information
  - Practical sessions: Python programming, Bachelor level, Year 1
- June–August 2025 **Internship supervision**, *3 months*, Samuel Godlewski, third-year Bachelor student  
in Applied Mathematics, Paris Dauphine University  
Internship topic: Security analysis of a variant of the Wave digital signature scheme

## Education

- March–August 2020 **Master's research internship**, *Inria*, Paris  
Topic: Quantum sieving for lattices
- 2018–2020 **Master's Degree in Applied Mathematics**, *University of Limoges*,  
Master's thesis: Quantum enumeration for the Shortest Vector Problem
- 2017–2018 **First year of Engineering School in Cybersecurity**, *INSA*, Bourges

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## Academic activities

2025 **Program Committee of PKC'26**

**External reviewer**, for *Eurocrypt 2023 et 2025, ANTS 2024, CRYPTO 2025 and 2026*

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## Publications

CiC 2026 **Quantum security analysis of Wave**, Johanna Loyer

CRYPTO'25 **Wagner's Algorithm Provably Runs in Subexponential Time for  $SIS^\infty$** , Léo Ducas, Lynn Engelberts, Johanna Loyer

CiC 2025 **Lattice Reduction via Dense Sublattices: A Cryptanalytic No-Go**, Léo Ducas, Johanna Loyer

DCC 2025 **Quantum sieving for code-based cryptanalysis and its limitations for ISD**, Lynn Engelberts, Simona Etinski, Johanna Loyer

PQCrypto'23 **Classical and Quantum 3 and 4-Sieves to Solve SVP with Low Memory**, André Chailloux, Johanna Loyer

AsiaCrypt'21 **Lattice Sieving via Quantum Random Walks**, André Chailloux, Johanna Loyer

[Submitted paper](#)

2025 **A Tight Quantum Algorithm for Multiple Collision Search**, Xavier Bonnetain, Johanna Loyer, André Schrottenloher, Yixin Shen

[Specification document](#)

2023 **Wave digital signature scheme**, *submitted to the NIST standardization process*, with G. Banegas, K. Carrier, A. Chailloux, A. Couvreur, T. Debris-Alazard, P. Gaborit, P. Karpman, R. Niederhagen, N. Sendrier, B. Smith, J-P. Tillich.



[Ph.D. Thesis Manuscript](#)

2023 **Quantum Cryptanalysis on Lattices and Codes**, *Inria Paris*, advised by André Chailloux and Nicolas Sendrier

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## Presentations

[International conferences with reading committee](#)

2023 **Post-Quantum Cryptography Conference (PQCrypto)**, *University of Maryland, College Park*, Classical and Quantum 3 and 4-Sieves to Solve SVP with Low Memory  
<https://pqcrypto2023.umiaccs.io/slides/index.html>

2021 **International Conference on the Theory and Application of Cryptology and Information Security (ASIACRYPT)**, *Lattice sieving via quantum walks*  
<https://asiacrypt.iacr.org/2021/acceptedpapers.php>

[International workshops on invitation](#)

2025 **Dagstuhl Seminar 25431**, *Leibniz Center for Informatics*, Schloss Dagstuhl (Allemagne)  
<https://www.dagstuhl.de/25431>

2023 **Dagstuhl Seminar 23421**, *Leibniz Center for Informatics*, Schloss Dagstuhl (Allemagne), Animation d'un groupe de travail en cryptanalyse des codes  
<https://www.dagstuhl.de/23421>

2022 **Quantum Cryptanalysis Workshop**, *Quantum Information Institute*, University of Bristol (Royaume-Uni), Sieving algorithms for cryptanalysis of lattices

#### Invited seminars

2025 **King's College**, *London (United Kingdom)*, Wagner's Algorithm Provably Runs in Subexponential Time for  $SIS^\infty$   
<https://jcs.trusted-third-party.org/d/2025-12-08>

**Cryptography Seminar**, *Institut de recherche mathématique de Rennes (IRMAR)*, Wagner's Algorithm Provably Runs in Subexponential Time for  $SIS^\infty$   
<https://www.creachlabs.fr/fr/node/802>

**GRACE team**, *LIX*, Palaiseau, Sieving attacks on lattice and code-based cryptography

2024 **Codes & Cryptographie**, *Inria*, Paris, Wagner's Algorithm Provably Runs in Subexponential Time for  $SIS^\infty$

**Cryptology group**, *CWI*, Amsterdam, Wagner's Algorithm Provably Runs in Subexponential Time for  $SIS^\infty$

2023 **Center for Quantum Technologies**, *National University of Singapore*, Lattice sieving via quantum walks

**Cryptology group**, *CWI*, Amsterdam, Classical and Quantum 3 and 4-Sieves to Solve SVP with Low Memory

**Codes & Cryptographie**, *ENS*, Lyon, Classical and Quantum 3 and 4-Sieves to Solve SVP with Low Memory

2022 **Joint seminar**, *ENS Lyon, CWI, KCL*, Lattice sieving via quantum walks  
<https://jcs.trusted-third-party.org/d/2022-01-10>

2021 **Cryptology group**, *CWI*, Amsterdam, Lattice sieving via quantum walks

**Codes & Cryptographie**, *Inria*, Paris, Lattice sieving via quantum walks

**COSMIQ team**, *Inria*, Paris, Lattice sieving via quantum walks

#### National workshops

2026 **Codes & Cryptographie days**, *Blériot*, Chair for the session on lattices

2022 **Codes & Cryptographie days**, *Hendaye*, Lattice sieving via quantum walks

#### Tutorials

2026 **GODZILLA Seminar**, *LIX*, Palaiseau, Decoding Algorithms

2025 **GODZILLA Seminar**, *LIX*, Palaiseau, Security analyses of lattice-based cryptography

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## Scientific Outreach

- 2025–2026 **Member of EclairSI association**, Association of scientists contributing to informed public debate on computer security and privacy
- 2024 **Interview in the popular science magazine *Epsilon* (n°38, July 2024)**
- 2023 **Organization of an outreach event**, *50 participants*, at the end of my Ph.D., aimed at raising awareness of cybersecurity, privacy, and the scientific method.
- 2021 **Participation in RJMI**, Animation of speed-meeting sessions with high school girls to promote scientific careers.
- 2018–2020 **Outreach talks in middle and high schools**, on prime numbers (in curriculum) and their use in cryptography
- 2018–2019 **Coordination of outreach talks on mathematics and post-high school orientation**, Conception and animation of sessions for high school classes on real-world use of mathematics and its importance in educational orientation. Teaching materials were made available to teachers.
- 2018 **Playful workshops in robotics and programming (Scratch, Ozobot)**, animation in primary school, in partnership with the INSPE of Bourges.

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## Languages

French native  
English fluent

SageMath Expert  
C Advanced